

GOCHANG, Korea, Republic of

WMO: 471720

Lat: 35.3489N

Lon: 126.5989E

Elev: 56

StdP: 100.65

Time Zone: 9.00 (E09)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.6	-6.9	-15.2	1.0	-3.3	-13.2	1.2	-2.7	6.0	-1.0	5.4	-0.9	1.6	0	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.2	33.1	25.7	32.0	25.5	30.7	24.8	27.2	30.2	26.4	29.8	25.8	29.3	2.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.5	22.2	28.9	25.6	20.9	28.3	24.8	20.0	27.9	86.5	30.0	83.0	29.9	80.2	29.5	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.0	5.2	4.5	-10.6	35.1	2.1	1.2	-12.1	35.9	-13.4	36.6	-14.5	37.3	-16.0	38.1		
			-11.4	28.0	2.0	1.4	-12.8	29.0	-13.9	29.9	-15.0	30.7	-16.5	31.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.8	-0.2	2.3	6.9	12.4	18.4	22.7	26.4	26.8	22.1	15.8	9.3	2.3	(6)
(7)		DBStd	9.82	3.73	4.01	3.71	3.57	2.69	1.80	2.16	2.42	2.50	3.09	4.12	3.97	(7)
(8)		HDD10.0	966	316	217	112	18	0	0	0	0	2	62	239		(8)
(9)		HDD18.3	2446	574	448	354	179	33	0	0	2	88	271	496		(9)
(10)		CDD10.0	2368	0	3	16	90	259	381	509	522	363	182	42	2	(10)
(11)		CDD18.3	806	0	0	0	2	34	131	251	264	114	9	1	0	(11)
(12)		CDH23.3	6696	0	0	2	17	278	763	2299	2627	671	39	0	0	(12)
(13)		CDH26.7	2048	0	0	0	1	38	144	738	995	130	2	0	0	(13)
(14)	Wind	WSAvg	1.7	1.6	1.8	2.0	2.0	1.8	1.7	1.9	1.7	1.5	1.4	1.7	1.8	(14)
(15)	Precipitation	PrecAvg	1352	33	43	61	86	94	156	286	299	143	54	55	40	(15)
(16)		PrecMax	1912	84	123	126	232	196	354	588	543	516	190	154	74	(16)
(17)		PrecMin	803	9	9	15	18	26	28	43	84	22	5	6	3	(17)
(18)		PrecStd	269	17	27	30	50	40	86	121	130	113	43	37	19	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.7	17.8	21.5	25.2	29.3	30.7	34.0	35.1	31.1	26.4	22.5	15.9	(19)
(20)		MCWB	8.6	13.7	12.5	14.5	17.1	20.6	25.8	25.9	23.9	20.7	17.1	12.0	(20)	
(21)		2%	DB	10.3	14.4	18.3	23.2	27.7	29.4	32.7	33.8	29.6	24.2	19.6	12.5	(21)
(22)		MCWB	6.8	10.9	11.3	14.3	17.0	20.6	25.8	25.6	23.1	17.7	15.2	9.3	(22)	
(23)		5%	DB	7.6	11.3	16.3	21.3	26.2	28.3	31.6	32.7	28.1	22.8	17.7	10.4	(23)
(24)		MCWB	4.4	7.0	10.9	13.0	16.6	20.5	25.5	25.6	22.1	16.9	14.0	7.6	(24)	
(25)		10%	DB	5.6	9.0	14.3	19.3	24.9	27.3	30.4	31.5	26.9	21.3	16.2	8.5	(25)
(26)		MCWB	2.7	5.6	9.5	12.4	16.3	20.3	25.1	25.5	21.3	16.4	12.8	5.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	14.8	15.0	17.7	21.9	24.3	28.7	28.2	25.7	22.9	18.1	13.2	(27)
(28)		MCDWB	11.6	16.6	17.7	22.4	23.7	27.0	31.1	31.5	29.1	24.9	21.3	14.4	(28)	
(29)		2%	WB	7.1	11.3	13.3	15.9	19.9	23.5	27.4	27.3	24.6	20.6	16.6	9.6	(29)
(30)		MCDWB	10.0	13.2	16.6	20.3	24.0	26.5	29.8	30.7	27.6	22.6	18.5	11.8	(30)	
(31)		5%	WB	4.8	7.8	11.6	14.8	18.7	22.7	26.4	26.7	23.6	19.0	14.9	7.9	(31)
(32)		MCDWB	6.9	10.5	15.6	18.7	23.4	25.8	29.3	30.2	26.5	20.6	17.4	10.0	(32)	
(33)		10%	WB	3.0	5.7	9.6	13.8	17.7	21.9	25.7	26.1	22.5	17.4	13.1	6.1	(33)
(34)		MCDWB	5.4	8.8	13.7	17.6	22.4	25.0	29.0	29.8	25.7	20.1	15.6	8.4	(34)	
(35)	Mean Daily Temperature Range		MDBR	7.8	8.5	10.5	11.4	11.3	8.9	6.6	7.2	8.6	10.0	8.7	7.5	(35)
(36)		5% DB	MCDBR	10.8	12.2	13.7	14.6	13.7	10.8	8.4	9.2	9.7	11.5	10.1	9.9	(36)
(37)		MCWBR	7.5	8.2	8.0	7.2	5.8	4.1	3.1	3.1	4.0	5.4	6.2	7.3	(37)	
(38)		5% WB	MCDWB	9.5	10.1	11.2	9.8	10.0	7.1	6.7	7.6	7.7	7.8	8.2	9.1	(38)
(39)		MCWBR	7.5	8.2	7.8	6.0	5.5	3.8	3.2	3.3	3.8	5.2	6.1	7.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.454	0.492	0.562	0.580	0.597	0.664	0.617	0.564	0.500	0.478	0.465	0.454	(40)	
(41)		taud	1.928	1.869	1.731	1.740	1.754	1.690	1.846	1.982	2.088	2.009	1.977	1.937	(41)	
(42)		Ebn at Noon	699	724	711	728	723	674	703	733	761	731	684	665	(42)	
(43)		Edh at Noon	147	175	218	227	227	242	205	176	151	150	139	138	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.33	3.17	4.35	5.06	5.54	5.01	4.57	4.68	4.21	3.69	2.54	2.04	(44)	
(45)		RadStd	0.19	0.29	0.39	0.40	0.59	0.45	0.61	0.67	0.40	0.31	0.28	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page